Scalability & Future Plan

Date	03 October 2026
Team ID	2
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	SQLite single-file database	Limits heavy concurrent writes and multi-server deployment	Medium
2	Single shared admin password; no end-user accounts	Users can't log back in to their plans from another device; one admin role only	High
3	Dependence on Gemini availability and free-tier quota	13–20 s responses or failures when the primary model is overloaded	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Uvicorn process; 50 users at 0% errors	Multiple workers behind a load balancer; deploy on Google Cloud Run with autoscaling
2	Data Storage	SQLite (WAL mode)	Migrate to PostgreSQL (Cloud SQL) via SQLAlchemy with Alembic migrations
3	Performance	Synchronous Gemini call per request (4–15 s)	Background job queue with progress updates; cache tips per goal; paid tier for higher limits
4	Security	Admin password + signed cookie; .env secrets	User accounts with OAuth (Google sign-in), role-based access for coaches, rate limiting, secrets in Secret Manager

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User accounts, cloud deployment (Cloud Run + Cloud SQL), PDF/calendar export	1–2 months	Users keep plans across devices; public access
Phase 3	Progress tracking and weekly check-ins that auto-adjust the next week's plan; exercise demo videos	3–4 months	Better adherence and long-term personalization
Phase 4	Coach multi-client workspace, wearable integration (steps, heart rate), multilingual UI (e.g. Tamil)	6 months	Supports gyms and institutions; wider reach